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ADAPTIVE DIODE EMULATION MODE OFFSET CIRCUIT

Abstract

An apparatus and method for an adaptive diode emulation mode offset circuit is disclosed. The apparatus includes a polarity detection circuit to identify the polarity of a phase node of a DC-DC converter; and an adaptive diode emulation mode (DEM) offset circuit to: detect a first early termination or a first late termination of an on time of a low-side gate drive of the DC-DC converter based on the identified polarity of the phase node; and output an offset voltage to selectively terminate on time of the low-side gate drive to mitigate a second early termination or a second late termination of the low-side gate drive, a value of the offset voltage based on a polarity of the phase node of the DC-DC converter; wherein the second early termination occurs after the first early termination and the second late termination occurs after the first late termination.

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Background/Summary

PRIORITY [0001] This application claims priority to U.S. Provisional Patent Application No. 63/558,252, filed Feb. 27, 2024, the contents of which are hereby incorporated in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a diode emulation mode circuit, specifically a diode emulation mode circuit with improved light-load efficiency.

BACKGROUND

[0003] Switch mode power supplies or switching regulators, also referred to as DC to DC converters, are often used to convert an input supply voltage to a desired output voltage. For example, a 5 volts supply voltage provided to an integrated circuit may need to be reduced to 2.8 volts to operate additional circuitry. A switching regulator provides power supply function through low loss components such as capacitors, inductors, and transformers, and power switches that are turned on and off to transfer energy from the input to the output in discrete packets. A feedback control circuit is used to regulate the energy transfer to maintain a constant output voltage within the desired load limits of the circuit.

[0004] A switching regulator can be configured to step up the input voltage or step down the input voltage or both. Specifically, a buck switching regulator, also called a “buck converter,” steps down the input voltage while a boost switching regulator, also called a “boost converter,” steps up the input voltage. A buck-boost switching regulator, or buck-boost converter, provides both step-up and step-down functions.

[0005] A conventional buck switching regulator includes a pair of power switches which are turned on and off to regulate an output voltage to be proportional to a reference voltage. More specifically, the power switches are alternately turned on and off to generate a switching output voltage at a switching output node, also referred to as the phase node. The phase node is coupled to an LC filter circuit including an output inductor and an output capacitor to generate an output voltage having substantially constant magnitude. The output voltage can then be used to drive a load.

[0006] In particular, the pair of power switches is often referred to as including a “high-side power switch” and a “low-side power switch.” The high-side power switch is turned on, or closed, to apply energy to the output inductor of the output filter circuit to allow the current through the inductor to build up. When the high-side power switch is turned off, or opened, the voltage across the inductor reverses and the current through the inductor reduces during this period.

[0007] The low-side power switch may be a diode, but as output voltages from DC-to-DC converters have gone down in magnitude, the low-side power switch may be replaced with a transistor to improve efficiency. However, unlike a diode, a transistor, such as a field effect transistor (FET) is a bi-directional switch, allowing current to flow in either direction. When the DC-to-DC converter operates under light current loads, the current in the inductor may fall to zero and even reverse. Reversed current reduces the efficiency of the DC-to-DC converter. To prevent reversal of current flow, diode emulation mode (DEM) circuitry may be added to detect when the inductor current goes to zero at which point the DEM circuitry turns the transistor off such that reverse current cannot flow.

[0008] Timing of the DEM circuitry is challenging and may be affected by many system parameters including, for example, the current ramp rate of the inductor, offset and propagation delay of the comparator in the DEM circuit, gate drive path propagation delay, operating temperature, and part-to-part variation. The DEM circuitry turning the low-side power switch off early, or late, can result in increased power dissipation, decreased efficiency, and performance

differences between parts. Thus, there is a need for DEM circuitry that can achieve a more accurate low-side power switch termination time.

SUMMARY OF THE INVENTION

[0009] Systems and methods for an adaptive diode emulation mode offset circuit are disclosed. The apparatus includes a polarity detection circuit to identify the polarity of a phase node of a DC-DC converter; and an adaptive diode emulation mode (DEM) offset circuit to: detect a first early termination or a first late termination of an on time of a low-side gate drive of the DC-DC converter based on the identified polarity of the phase node; and output an offset voltage to selectively terminate on time of the low-side gate drive to mitigate a second early termination or a second late termination of the low-side gate drive, a value of the offset voltage based on a polarity of the phase node of the DC-DC converter; wherein the second early termination occurs after the first early termination and the second late termination occurs after the first late termination.

[0010] An apparatus includes a DC-DC converter including a low-side gate drive and a phase node, a polarity detection circuit to identify a polarity of the phase node, an adaptive diode emulation mode (DEM) offset circuit, and a DEM circuit. The adaptive DEM offset circuit to: receive the identified polarity of the phase node from the polarity detection circuit; detect a first early termination or a first late termination of an on time of the low-side gate drive based on the identified polarity of the phase node; and output an offset voltage, a value of the offset voltage based on the polarity of the phase node. The DEM circuit to: receive the offset voltage; and selectively terminate the on time of the low-side gate drive to mitigate a second early termination or a second late termination of the low-side gate drive.

[0011] A method includes receiving a polarity of a phase node of a DC-DC converter; detecting a first early termination or a first late termination of an on time of a low-side gate drive of the DC-DC converter based on the received polarity of the phase node; and outputting an offset voltage to selectively terminate the on time of the low-side gate drive to mitigate a second early termination or a second late termination of the on time of the low-side gate drive, a value of the offset voltage based on a polarity of a phase node.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The figures illustrate examples of systems and methods.

[0013] FIG. 1 illustrates a system including an adaptive DEM circuit in a DC-DC converter, according to examples of the present disclosure;

[0014] FIG. 2 illustrates an DC-to-DC converter, according to examples of the present disclosure;

[0015] FIG. 3 illustrates a polarity detection circuit, according to examples of the present disclosure;

[0016] FIG. 4 illustrates an adaptive DEM offset circuit, according to examples of the present disclosure;

[0017] FIG. 5 illustrates a DEM circuit, according to examples of the present disclosure;

[0018] FIG. 6 illustrates simulation results of a DC-to-DC converter containing an adaptive DEM offset circuit, according to examples of the present disclosure;

[0019] FIG. 7 illustrates a method performed for using an adaptive DEM circuit in a DC-DC converter, according to examples of the present disclosure; and

[0020] FIG. 8 illustrates a more detailed version of the method described in FIG. 7, according to examples of the present disclosure.

[0021] The reference number for any illustrated element that appears in multiple different figures has the same meaning across the multiple figures, and the mention or discussion herein of any illustrated element in the context of any particular figure also applies to each other figure, if any, in

which that same illustrated element is shown.

DESCRIPTION

[0022] According to an aspect of the invention, an adaptive diode emulation mode (DEM) offset circuit is provided. The DEM offset circuit achieves more accurate timing for terminating a switch in a direct current (DC) to DC converter to improve the efficiency of the DC-to-DC converter.

[0023] FIG. 1 illustrates a system including an adaptive DEM circuit in a DC-to-DC converter, according to examples of the present disclosure. System **100** may include DC-to-DC converter **110**, polarity detection circuit **120**, adaptive DEM offset circuit **130**, and DEM circuit **140**.

[0024] DC-to-DC converter **110** may be a circuit to convert direct current from one voltage to another voltage, such as a step-down (buck) converter, step-up (boost) converter, or a buck-boost converter. DC-to-DC converter **110** may be connected to an input voltage ($V_{sub.IN}$) and ground (GND), or other return. DC-to-DC converter **110** may include two gate drives (e.g., a high-side gate drive and a low-side gate drive) and at least one energy storage element, such as a capacitor, an inductor, or any combination thereof. In some examples, one of the gate drives may be a transistor, such as a field effect transistor (FET). The state of the low-side gate drive in DC-to-DC converter **110** may be controlled by DEM circuit **140** to time the termination of the on time of the low-side gate drive to improve the efficiency of DC-to-DC converter **110**. A more detailed example of DC-to-DC converter **110** is shown in FIG. 2.

[0025] Polarity detection circuit **120** may be a circuit to determine the polarity of a phase node in DC-to-DC converter **110**. Polarity detection circuit **120** may receive the phase node polarity, or a representative thereof, and may compare the phase node voltage against a reference voltage to determine whether the termination of the on time of the low-side gate drive in DC-to-DC converter **110** is terminated early (e.g., the phase node voltage is below ground) or late (e.g., the phase node voltage is above $V_{sub.IN}$). Polarity detection circuit **120** may output the polarity of the phase node (e.g., positive or negative) to adaptive DEM offset circuit **130**. A more detailed example of polarity detection circuit **120** is shown in FIG. 3.

[0026] Adaptive DEM offset circuit **130** may be a circuit to detect whether the low-side gate drive in DC-to-DC converter **110** may have been terminated early or late. For example, if the polarity of the phase node is positive, the low-side gate drive may have been terminated late and if the polarity of the phase node is negative, the low-side gate drive may have been terminated early. Adaptive DEM offset circuit **130** may also output an offset voltage based on the polarity of the phase node. The offset voltage may be output to DEM circuit **140**. A more detailed example of adaptive DEM offset circuit **130** is shown in FIG. 4.

[0027] DEM circuit **140** may be a circuit to receive the offset voltage from adaptive DEM offset circuit **130** and output a control signal to control the state of the low-side gate drive to DC-to-DC converter **110**. The control signal may be used to selectively terminate the on time of the low-side gate drive to more accurately time the termination of the on time of the low-side gate drive to mitigate a subsequent early or late termination. In some examples, the control signal output by DEM circuit **140** may be a logic value input to the low-side gate drive (e.g., a logic value of 1 turns the low-side gate drive on and a logic value of 0 turns the low-side gate drive off). A more detailed example of DEM circuit **140** is shown in FIG. 5.

[0028] FIG. 2 illustrates an DC-to-DC converter, according to examples of the present disclosure. In the example shown in FIG. 2, the DC-to-DC converter is a step-down or buck converter. DC-to-DC converter **210** is illustrated in FIG. 2 as including four main components: high-side power switch **211**, low-side switch **212**, inductor **214**, and capacitor **215**.

[0029] Low-side switch **212**, which may be a transistor such as a FET or metal-oxide-semiconductor field-effect transistor (MOSFET), may be controlled by an input logic value, labelled "LG" in FIG. 2, from a DEM circuit, such as DEM circuit **140** shown in FIG. 1. If the on time of low-side switch **212** is terminated late, the current direction through inductor **214** may reverse, causing the voltage at phase node **216** to rapidly increase to above $V_{sub.IN}$ after

termination of low-side switch **212**, resulting in increased power dissipation. If the on time of low-side switch **212** is terminated early, the voltage at phase node **216** is pulled below ground by the current circulating through DC-to-DC converter **210**. Specifically, because the current is flowing from ground to phase node **216** through low-side switch **212** and ground is a zero-volt reference, the voltage at phase node **216** will be pulled below ground for current to flow. Early termination also results in increased power dissipation over an accurately timed termination of low-side switch **212**. The offset voltage from a DEM offset circuit, such as adaptive DEM offset circuit **130**, may provide an offset voltage to the DEM circuit, such as DEM circuit **140**, to achieve a more accurately timed termination of low-side switch **212** and mitigate the occurrence of early or late terminations of the on time of low-side switch **212**.

[0030] DC-to-DC converter **210** may also include high-side power switch **211** which may be a second switch, such as a diode or transistor, and may be used to selectively provide the input voltage to DC-to-DC converter **210**. The position of high-side power switch **211** may be controlled by input logic value labeled “HG” in FIG. 2. Capacitor **215** may be a capacitor that works in conjunction with inductor **214** to store energy. Capacitor **215** may have any suitable value, such as 1 mF.

[0031] DC-to-DC converter **210** is also illustrated as including error amplifier **213**, current sensor **221**, pulse width modulation (PWM) comparator **217**, set-reset (SR) latch **218**, inverter **220**, and clock **219** which may be used to close the power supply loop and provide a fixed output voltage, $V_{sub.OUT}$. Error amplifier **213** may compare the output voltage to a reference voltage. Current sensor **221** may sense the current of inductor **214**. PWM comparator **217** may compare the signal from current sensor **221** with the output of error amplifier **213**. The output of PWM comparator **217** may be provided to SR latch **218** and used to reset SR latch **218**. SR latch **218** may latch the high-side gate drive signal (labelled “HG”) to high at every clock cycle and to low with each reset from PWM comparator **217**. Inverter **220** may ensure that the low-side gate drive signal (labelled “LG”) is maintained at an opposite state of the high-side gate drive signal by outputting a signal (labelled “nHG”) that is opposite of the high-side gate drive signal. Clock **219** may provide a signal to SR latch **218** to signal the start of each switching cycle. Clock **219** may also provide a signal to fix the PWM frequency.

[0032] To implement DEM in a DC-to-DC converter, such as DC-to-DC converter **210**, DEM comparator circuit **140** may be added to DC-to-DC converter **210**. AND gate **222** may receive the output of inverter **220** and the output of DEM circuit **140** as inputs and output the low-side gate drive signal, LG. The output of DEM circuit **140** may occur earlier than if DEM comparator circuit **140** were not present (e.g., if the output of PWM comparator **217** were used to provide the low-side gate drive signal).

[0033] FIG. 3 illustrates a polarity detection circuit, according to examples of the present disclosure. Polarity detection circuit **320** may receive the voltage of the phase node of the DC-to-DC converter, such as phase node **216** shown in FIG. 2. Polarity detection circuit **320** may determine the polarity of the phase node based on the received voltage. Specifically, comparator **322** may compare the voltage of the phase node against a direct current reference voltage, labelled $V_{sub.PREF}$ in FIG. 3. The DC reference voltage may be any reference voltage, greater than ground and less than or equal to $V_{sub.IN}$, in the system of which the DC-to-DC converter is a part. For example, $V_{sub.IN}$ may be selected as the reference voltage for DEM polarity detection circuit **320** because even if the voltage of the phase node oscillates after the synchronous switch is terminated, in the event of a late termination, the voltage at the phase node will rise above $V_{sub.IN}$ by the forward voltage drop of the high-side switch (e.g., the forward voltage drop of the body diode of the high-side switch) and in the event of a late termination, the phase node voltage will be less than ground, and the voltage of the phase node will not exceed $V_{sub.IN}$. The polarity of the phase node immediately after the on time of the low-side gate drive is terminated may be used to determine if the termination of the on time occurred early (e.g., the phase node voltage is below

ground) or late (e.g., the phase node voltage is above $V_{sub.IN}$).

[0034] Polarity detection circuit **320** may include flip flop switch **324**. Flip flop switch **324** may sample the output of comparator **322** (e.g., the polarity of the phase node) when the clock input sees a rising edge, provided by a delayed DEM OUT signal. The DEM OUT signal may be from a DEM circuit, such as DEM circuit **140** shown in FIG. 2 and described in more detail with respect to FIG. 5. Delay **323** may provide a short delay on the DEM signal from a DEM circuit, such as DEM circuit **140** shown in FIG. 1, labelled “DEM OUT” in FIG. 3. The delay provided by delay **323** may ensure that the sampling performed by flip flop switch **324** occurs slightly after the termination of the low-side gate drive, such as low side gate drive **212** shown in FIG. 2.

[0035] The output of flip flop switch **324** is determined by the polarity of the phase node during the current cycle of the clock, such as clock **219** shown in FIG. 2. Specifically, when the polarity of the phase node is positive (e.g., the voltage of the phase node is greater than the reference voltage), the output of Q of flip flop switch **324** may be one (1) and when the polarity of the phase node is negative (e.g., the voltage of the phase node is less than the reference voltage), the output of Q flip flop switch **324** may be one (1). In FIG. 3, an output of Q is labeled “POL” and an output of Q is labeled “nPOL.” The outputs of flip flop switch **324** (e.g., POL and nPOL) may be provided as inputs to an adaptive DEM offset circuit, such as adaptive DEM offset circuit **130** shown in FIG. 2 and as described in more detail with respect to FIG. 4.

[0036] FIG. 4 illustrates an adaptive DEM offset circuit, according to examples of the present disclosure. Adaptive DEM offset circuit **430** may receive the polarity signal from a polarity detection circuit, such as polarity detection circuit **320** shown in FIG. 3, and the current DEM OUT signal from DEM circuit **140**. The DEM OUT signal may indicate whether the current in the inductor, such as inductor **214** shown in FIG. 2, is crossing zero. The polarity signal may be added to the DEM OUT signal at AND gates **431a** and **431b**. The output of AND gates **431a** and **431b** may be input to one shot timers **432a** and **432b**. One shot timers **432a** and **432b** may control the charge and discharge duration of capacitor **434**. Specifically, one shot timer **432a** outputs a logic value of 1 (labeled “CHG” in FIG. 4) when the polarity of the phase node is negative and one shot timer **432b** outputs a logic value of 1 (labeled “DIS” in FIG. 4) when the polarity of the phase node is positive.

[0037] Adaptive DEM offset circuit **430** may receive current from current sources **433a** and **433b**. The size of current sources **433a** and **433b** may be any suitable amount. In some examples, current source **433a** may have a different size than current source **433b**. At switch **435a**, when one shot timer **432a** outputs CHG, switch **435a** closes to provide current from current source **433a** to capacitor **434** to charge capacitor **434**. At switch **435b**, when one shot timer **432b** outputs DIS, switch **435b** closes to provide the current from current source **433b** to capacitor **434** to discharge capacitor **434**. Thus the voltage level of capacitor **434** is raised and lowered by the opening and closing of switches **435a** and **435b**. The voltage level of capacitor **434** is buffered at buffer **436** and output as an offset voltage, labeled “DEM REF” in FIG. 4. The offset voltage, DEM REF, may be the reference for a DEM circuit, such as DEM circuit **140** shown in FIG. 2 and as described in more detail with respect to FIG. 5. The offset voltage may be forced to zero millivolts when the current through the inductor of the DC-to-DC converter, such as inductor **214** shown in FIG. 2, is not crossing zero.

[0038] The period of one-shot timers **432a** and **432b**, the strength of current sources **433a** and **433b**, and the value of capacitor **434** may determine the step size of the adjustment of DEM REF. These values may be adjusted based on the implementation of the adaptive DEM offset circuit and the DC-to-DC converter into which the adaptive DEM offset circuit is incorporated to provide the optimal step size and die area for the application.

[0039] FIG. 5 illustrates a DEM circuit, according to examples of the present disclosure. DEM circuit **540** may receive the voltage at the phase node (“PHASE”) and the offset voltage (“DEM REF”) from an adaptive DEM offset circuit. Comparator **542** compares the voltage at the phase

node and the offset voltage and outputs a “1” if the voltage at the phase node is greater than the offset voltage or a “0” if the offset voltage is greater than the voltage at the phase node. The output of comparator **542** is added the inverted high-side gate drive signal, nHG, as described with respect to FIG. 2 above, at AND gate **544**. The output of AND gate **544** is then provided to SR latch **546**. SR latch **546** may be set when the high-side gate drive is low. SR latch **546** may be reset by the high-side gate drive signal, HG, every switching cycle to ensure that diode emulation mode is checked independently each cycle. SR latch **546** may output a logic value labeled “DEM OUT” in FIG. 5 indicating whether the low side gate drive in the DC-to-DC converter should be opened or closed. DEM OUT may be provided to the components of the DC-to-DC converter, such as AND gate **222** shown in FIG. 2, to control the position of the low-side gate drive.

[0040] DEM circuit **540** may be similar to a standard DEM circuit with the addition of the output of the adaptive DEM offset circuit, DEM REF. DEM REF may replace the reference value used in a standard DEM circuit, which may be a fixed value.

[0041] FIG. 6 illustrates simulation results of a DC-to-DC converter containing an adaptive DEM offset circuit, according to examples of the present disclosure. Graph **600** illustrates the current through the inductor of the DC-to-DC converter, such as inductor **214** shown in FIG. 2, in area **630** of graph **600**. When the current through the inductor is not at or crossing zero, the offset voltage to the DEM comparator (labeled DEM_REF in FIG. 6) is forced to 0 mV. In the example shown in graph **600**, at approximately 1.42 milliseconds, the current through the inductor reaches zero. At that time, the offset voltage (DEM REF) begins to be adjusted in steps until the polarity at the phase node changes. While in the example shown in area **620** of graph **600**, the offset voltage is adjusted in 100 μ V increments until the polarity of the phase node changes, in other examples, the increments may be any size.

[0042] Graph **600** illustrates the output from a polarity detection circuit, such as polarity detection circuit **320** shown in FIG. 3, in area **610** of graph **600**. At approximately 1.495 milliseconds, the polarity of the phase node reverses. Once the polarity of the phase node reverses, the offset voltage increases until the polarity of the phase node reverses again. During steady state operation, the offset voltage may continue to increase and decrease between positive and negative polarity of the phase node to achieve accurate timing of the termination of the low-side gate drive in the DC-to-DC circuit, such as low-side switch **212** shown in FIG. 2.

[0043] FIG. 7 illustrates a method performed for using an adaptive DEM circuit in a DC-DC converter, according to examples of the present disclosure. Method **700** may be implemented using an adaptive DEM offset circuit, such as adaptive DEM offset circuit **430** shown in FIG. 4, or any other circuit operable to implement method **700**. Although examples have been described above, other variations and examples may be made from this disclosure without departing from the spirit and scope of these disclosed examples.

[0044] Method **700** may begin at block **710** where an adaptive DEM offset circuit may receive a polarity of a phase node of a DC-DC converter. The polarity of the phase node may be output to the adaptive DEM offset circuit by a polarity detection circuit, such as polarity detection circuit **320** shown in FIG. 3.

[0045] At block **720**, the adaptive DEM offset circuit may detect a first early termination or a first late determination of an on time of a low-side gate drive in the DC-DC converter. The adaptive DEM offset circuit may detect the termination of the on time of the low-side gate drive and determine that the termination was early or late based on the polarity of the phase node immediately after the low-side gate drive on time is terminated. The termination may have occurred early when the phase node voltage is below ground or late when the phase node voltage is above $V_{sub.IN}$.

[0046] At block **740**, the adaptive DEM offset circuit may output an offset voltage to selectively terminate the on time of the low-side gate drive to mitigate a second early termination or a second late termination of the on time of the low-side gate drive. The adaptive DEM offset circuit may

output the offset voltage by selectively charging a capacitor in the adaptive DEM offset circuit, such as capacitor **434** shown in FIG. **4**. The offset voltage may be based on the polarity of the phase node received at block **710**.

[0047] Although FIG. **7** discloses a particular number of operations related to method **700**, method **700** may be executed with greater or fewer operations than those depicted in FIG. **7**. In addition, although FIG. **7** discloses a certain order of operations to be taken with respect to method **700**, the operations comprising method **700** may be completed in any suitable order.

[0048] FIG. **8** illustrates a more detailed version of the method described in FIG. **7**, according to examples of the present disclosure. Method **800** may be implemented using an adaptive DEM offset circuit, such as adaptive DEM offset circuit **430** shown in FIG. **4**, or any other circuit operable to implement method **800**. Although examples have been described above, other variations and examples may be made from this disclosure without departing from the spirit and scope of these disclosed examples.

[0049] Method **800** may begin at block **810** where an adaptive DEM offset circuit may receive a polarity of a phase node of a DC-DC converter. The polarity of the phase node may be output to the adaptive DEM offset circuit by a polarity detection circuit, such as polarity detection circuit **320** shown in FIG. **3**.

[0050] At block **820**, the adaptive DEM offset circuit may detect a first early termination or a first late determination of an on time of a low-side gate drive in the DC-DC converter. The adaptive DEM offset circuit may detect the termination of the on time of the low-side gate drive and determine that the termination was early or late based on the polarity of the phase node immediately after the low-side gate drive is terminated. The termination of the on time may have occurred early when the phase node voltage is below ground or late when the phase node voltage is above $V_{sub.IN}$.

[0051] At block **830**, the adaptive DEM offset circuit may dynamically adjust the offset voltage based on the polarity of the phase node after a previous termination of the on time of the low-side gate drive. The offset voltage may be dynamically adjusted by selectively charging a capacitor in the adaptive DEM offset circuit, such as capacitor **434** shown in FIG. **4**. For example, the polarity of the phase node may be based on the previous termination of the on time of the low-side gate drive. When the polarity of the phase node is negative (e.g., the on time of the low-side gate drive previously was terminated early), the capacitor may be charged and when the polarity of the phase node is positive (e.g., the on time of the low-side gate drive previously was terminated late), the capacitor may be discharged.

[0052] At block **835**, the adaptive DEM offset circuit may repeatedly adjust the offset voltage in increments until the polarity of the phase node reverses. For example, the capacitor in the adaptive DEM offset circuit may be charged in increments to create offset voltage increments. The offset voltage may be adjusted by these increments until the polarity of the phase node reverses as described in further detail with respect to FIG. **6**. The offset voltage increments may have any suitable size. In some examples, the increments may be of equal magnitude and in other examples, the increments may be of different magnitude.

[0053] At block **840**, the adaptive DEM offset circuit may output an offset voltage to selectively terminate the on time of the low-side gate drive to mitigate a second early termination or a second late termination of the on time of the low-side gate drive. The adaptive DEM offset circuit may output the offset voltage by selectively charging a capacitor in the adaptive DEM offset circuit, such as capacitor **434** shown in FIG. **4**. The offset voltage may be based on the polarity of the phase node received at block **810**.

[0054] Although FIG. **8** discloses a particular number of operations related to method **800**, method **800** may be executed with greater or fewer operations than those depicted in FIG. **8**. In addition, although FIG. **8** discloses a certain order of operations to be taken with respect to method **800**, the operations comprising method **800** may be completed in any suitable order.

[0055] While the adaptive DEM zero crossing detection is illustrated with reference to a buck converter, the adaptive DEM zero crossing detection circuitry may also be used in any DC-DC converter topology including buck-boost and synchronous boost topologies.

[0056] Although examples have been described above, other variations and examples may be made from this disclosure without departing from the spirit and scope of these disclosed examples.

Claims

1. An apparatus, comprising: a polarity detection circuit to identify the polarity of a phase node of a DC-DC converter; and an adaptive diode emulation mode (DEM) offset circuit to: detect a first early termination or a first late termination of an on time of a low-side gate drive of the DC-DC converter based on the identified polarity of the phase node; and output an offset voltage to selectively terminate on time of the low-side gate drive to mitigate a second early termination or a second late termination of the low-side gate drive, a value of the offset voltage based on a polarity of the phase node of the DC-DC converter; wherein the second early termination occurs after the first early termination and the second late termination occurs after the first late termination.
2. The apparatus of claim 1, wherein the adaptive DEM offset circuit is to dynamically adjust the offset voltage based on the polarity of the phase node.
3. The apparatus of claim 2, wherein the adaptive DEM offset circuit is to repeatedly adjust the offset voltage in increments until the polarity of the phase node reverses.
4. The apparatus of claim 1, wherein the adaptive DEM offset circuit is to receive a polarity of a phase node of a DC-DC converter.
5. The apparatus of claim 1, wherein the adaptive DEM offset circuit is to detect the first early termination through detection of a voltage of the phase node being pulled below ground.
6. The apparatus of claim 1, wherein the adaptive DEM offset circuit is to detect the first late termination through detection of a voltage of the phase node being pulled above the input voltage.
7. The apparatus of claim 1, wherein the offset voltage is approximately zero when a current through the DC-DC converter flows in a positive direction.
8. An apparatus, comprising: a DC-DC converter including a low-side gate drive and a phase node; a polarity detection circuit to identify a polarity of the phase node; an adaptive diode emulation mode (DEM) offset circuit to: receive the identified polarity of the phase node from the polarity detection circuit; detect a first early termination or a first late termination of an on time of the low-side gate drive based on the identified polarity of the phase node; and output an offset voltage, a value of the offset voltage based on the polarity of the phase node; and a diode emulation mode (DEM) circuit to: receive the offset voltage; and selectively terminate the on time of the low-side gate drive to mitigate a second early termination or a second late termination of the low-side gate drive.
9. The apparatus of claim 8, wherein the adaptive DEM offset circuit is to dynamically adjust the offset voltage based on the polarity of the phase node.
10. The apparatus of claim 9, wherein the adaptive DEM offset circuit is to repeatedly adjust the offset voltage in increments until the polarity of the phase node reverses.
11. The apparatus of claim 10, wherein the increments are of equal magnitude.
12. The apparatus of claim 8, wherein the adaptive DEM offset circuit is to detect the first early termination through detection of a voltage of the phase node being pulled below ground.
13. The apparatus of claim 8, wherein the adaptive DEM offset circuit is to detect the first late termination through detection of a voltage of the phase node being pulled above the input voltage.
14. The apparatus of claim 8, wherein the offset voltage is approximately zero when a current through the DC-DC converter is greater than zero.
15. A method, comprising: receiving a polarity of a phase node of a DC-DC converter; detecting a first early termination or a first late termination of an on time of a low-side gate drive of the DC-

DC converter based on the received polarity of the phase node; and outputting an offset voltage to selectively terminate the on time of the low-side gate drive to mitigate a second early termination or a second late termination of the on time of the low-side gate drive, a value of the offset voltage based on a polarity of a phase node.

16. The method of claim 15, comprising dynamically adjusting the offset voltage based on the polarity of the phase node after a previous termination of the low-side gate drive.

17. The method of claim 16, comprising repeatedly adjusting the offset voltage in increments until the polarity of the phase node reverses.

18. The method of claim 17, wherein the increments are of equal magnitude.

19. The method of claim 15, wherein detecting a first early termination includes detecting a voltage of the phase node being pulled below ground.

20. The method of claim 15, wherein detecting a first late termination includes detecting a voltage of the phase node being pulled above the input voltage.
